

Electricity and Magnetism — Lesson 1

ICSE Class 7 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Law of magnetism, repulsion as the sure test, magnetic field, electromagnet, making an electromagnet and its uses.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. Two unlike poles of magnets brought near each other will:

- | | |
|---------------|------------|
| A. do nothing | B. repel |
| C. melt | D. attract |

2. A magnet attracts an unmagnetised iron nail, but a magnet also ____ the like pole of another magnet.

- | | |
|-------------|------------|
| A. heats | B. ignores |
| C. attracts | D. repels |

3. The space around a magnet where its force can be felt is the:

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|-------------------|------------------|
| A. magnetic field | B. neutral point |
| C. magnetic pole | D. armature |

4. An electromagnet is made by winding insulated wire carrying current around a:

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|-------------------|---------------|
| A. plastic rod | B. copper rod |
| C. soft-iron core | D. glass tube |

5. The magnetism of an electromagnet disappears as soon as:

- | | |
|--------------------------------|-----------------|
| A. the current is switched off | B. it is moved |
| C. it is painted | D. it is cooled |

6. To make an electromagnet stronger, you can wind ____ turns of wire.

- | | |
|----------|-----------|
| A. fewer | B. more |
| C. no | D. looser |

7. A device that uses an electromagnet to lift heavy iron scrap is a:

- | | |
|-----------|------------|
| A. lever | B. crane |
| C. pulley | D. crowbar |

8. The poles of an electromagnet can be changed by reversing the:

- | | |
|--------------------|----------------------|
| A. number of turns | B. current direction |
| C. core | D. colour |

9. Which of these can an electromagnet, but not a permanent magnet, do?

- | | |
|---------------------------|--------------------------|
| A. be switched on and off | B. have a magnetic field |
| C. have two poles | D. attract iron |

10. When the like poles of two magnets are brought together, the force between them is one of:

- | | |
|---------------|-------------|
| A. attraction | B. no force |
| C. repulsion | D. gravity |

Answer Key

1. D

2. D

3. A

4. C

5. A

6. B

7. B

8. B

9. A

10. C